

On Regularization of Maximum Mean Discrepancy Gradient Flow

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Problem: How to learn a target probability distribution π on $\mathbb{R}^d$.

- Sampling (e.g: $\pi \propto \exp(-V)$ is the posterior distribution in Bayesian inference).
- Generative models (e.g: π is the distribution of an image dataset).

Background: Optimization in the space of probability measures

Problem: How to learn a target probability distribution π on $\mathbb{R}^d$.

- This problem can be written as an **optimization** problem on $\mathcal{P}_2(\mathbb{R}^d)$.

$$\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi).$$

- Here D is a similarity metric or distance, e.g. Kullback–Leibler divergence.
 - $D(\mu \mid \pi) = 0$ if and only if $\mu = \pi$.
- $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures with a finite second moment.
 - We primarily consider $\mathcal{P}_2(\mathbb{R}^d)$ rather than $\mathcal{P}(\mathbb{R}^d)$ for the nice geometrical properties.
- How to find the minimum? Gradient descent!

Background: Euclidean gradient flow

- Euclidean gradient flow of an objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\partial_t x_t = -v(x_t), \quad v = \nabla F.$$

- ∇F denotes the gradient of F .
- This is the continuous-time analogue of gradient descent:

$$x_{n+1} = x_n - \gamma v(x_n),$$

where $\gamma > 0$ is the step size.

- When F is both **strongly convex and smooth**, Euclidean gradient **descent** converges exponentially fast [BV04].

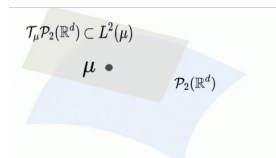
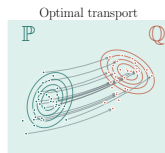
Background: Wasserstein gradient flow

Challenge: How to find $\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi)$?

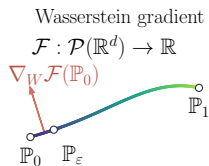
- Gradient descent! Wait... How do we define gradients in $\mathcal{P}_2(\mathbb{R}^d)$?
- Endow $\mathcal{P}_2(\mathbb{R}^d)$ with the Wasserstein-2 distance W_2 .

$$W_2^2(\nu, \mu) = \int \|T(x) - x\|^2 d\nu(x) = \|T - \text{Id}\|_{L^2(\nu)}^2.$$

- $W_2^2(\nu, \mu)$ means the **minimal energy** takes to transport mass from ν to μ .
- $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the **optimal transport map** from ν to μ .
- $(\mathcal{P}_2, W_2)$ can be 'treated' as a Riemann manifold under Otto's calculus [Ott01].
- The **tangent space** $\mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$ at $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is a dense subset of $L^2(\mu)$.



Background: Wasserstein gradient flow



Definition (Wasserstein gradient)

Let $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$. The **Wasserstein gradient of $\mathcal{F}$** evaluated at $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is the unique function $\nabla_{W_2} \mathcal{F}(\mu) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, s.t. for any $T \in \mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\mathcal{F}((\text{Id} + \epsilon T)_\# \mu) - \mathcal{F}(\mu)] = \langle \nabla_{W_2} \mathcal{F}, T \rangle_{L^2(\mu)}.$$

- The gradient is defined along a ‘curve’ $(\text{Id} + \epsilon T)_\# \mu$ in $\mathcal{P}_2(\mathbb{R}^d)$.

Background: Wasserstein gradient flow

Definition (Wasserstein gradient flow)

Wasserstein gradient flow of $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ is a sequence of distributions $(\mu_t)_{t \geq 0}$ that evolves according to the **continuity equation** $\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0$, where the vector field $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$. Or equivalently, $\dot{x}_t = v_t(x_t)$ with $\mu_t = \text{Law}(x_t)$.

Euclidean Gradient Flow

- State space: $\mathbb{R}^d$
- Objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$.
- Update scheme: $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla F$.

Wasserstein Gradient Flow

- State space: $\mathcal{P}_2(\mathbb{R}^d)$
- Objective $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$:
- Update scheme $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$.

Example 1: Langevin diffusion [JKO98]

- Given the target distribution $\pi \propto \exp(-V)$ with $V : \mathbb{R}^d \rightarrow \mathbb{R}$.
- The functional $\mathcal{F}_{\text{KL}} = \text{KL}(\cdot \| \pi)$ and its Wasserstein gradient

$$[\nabla_{W_2} \mathcal{F}_{\text{KL}}(\mu)](\cdot) = \nabla V(\cdot) + \nabla \log \mu_t(\cdot).$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

$$\partial_t \mu_t = \nabla \cdot (\mu_t (\nabla V + \nabla \log \mu_t)).$$

- It is equivalent to the Fokker–Planck equation of the **Langevin diffusion** [SS19]:

$$\mathrm{d}x_t = -\nabla V(x_t) \mathrm{d}t + \sqrt{2} \mathrm{d}W_t, \quad \mu_t = \text{Law}(x_t).$$

- A standard time-discretization (Euler–Maruyama scheme) is:

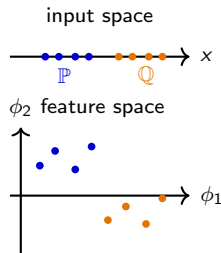
$$x_{n+1} = x_n - \gamma \nabla V(x_n) + \sqrt{2\gamma} \xi_n, \quad \xi_n \sim \mathcal{N}(0, I_d).$$

Background: Maximum Mean Discrepancy (MMD) [GBR⁺12]

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



MMD admits a closed-form expression

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian) [SFL11].

Example 2: MMD gradient flow [AKSG19]

- Given M i.i.d samples $\{y_i\}_{i=1}^M$ from a target distribution π .
- The functional $\mathcal{F}_{\text{MMD}} = \frac{1}{2}\text{MMD}^2(\cdot\|\pi)$
- Its Wasserstein gradient

$$\begin{aligned} [\nabla_{W_2}\mathcal{F}_{\text{MMD}}(\mu)](\cdot) &= \nabla \left(\int k(x, \cdot) \, \mathrm{d}\mu(x) - \int k(x, \cdot) \, \mathrm{d}\pi(x) \right) \\ &\approx \int \nabla_2 k(x, \cdot) \, \mathrm{d}\mu(x) - \frac{1}{M} \sum_{i=1}^M \nabla_2 k(y_i, \cdot). \end{aligned}$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$,

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)), \quad \mathrm{d}x_t = -[\nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)](x_t) \, \mathrm{d}t.$$

- A standard time-discretization (Euler–Maruyama scheme) is:

$$x_{n+1} = x_n - \gamma \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_n)(x_n), \quad \text{Law}(x_n) = \mu_n.$$

Convexity and Smoothness

Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

- $\pi \propto \exp(-V)$
- Sampling
- When $\pi \propto \exp(-V)$ is **strongly log-concave**, i.e., $\mathbf{H}V \geq \alpha \text{Id}$, then $\mathcal{F}_{\text{KL}}$ is **α -convex**.
- Convergence in discrete time

$$\mathcal{F}_{\text{KL}}(\mu_n \| \pi) \leq \exp(-\alpha\gamma n) \mathcal{F}_{\text{KL}}(\mu_0 \| \pi) + \frac{\gamma n \beta^2}{\alpha}.$$

- β is the Lipschitz continuity of V .
- It takes $\mathcal{O}(\frac{1}{\alpha\delta} \log \frac{1}{\delta})$ to reach δ error.

Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$

- $\{y_i\}_{i=1}^M$ i.i.d samples from π
- Generative modelling
- When k is bounded and has bounded derivatives, then $\mathcal{F}_{\text{MMD}}$ is **M -smooth and $-M$ -convex**.
- **Non-convergence** in discrete time

$$\mathcal{F}_{\text{MMD}}(\mu_n, \pi) \leq \frac{W_2^2(\mu_0, \pi)}{\gamma n} + \bar{K}.$$

- $\bar{K}$ is a positive barrier term that does not vanish.
- $\lim_{n \rightarrow \infty} \mathcal{F}_{\text{MMD}}(\mu_n, \pi) \neq 0!$

Non-convexity of MMD prevents global convergence

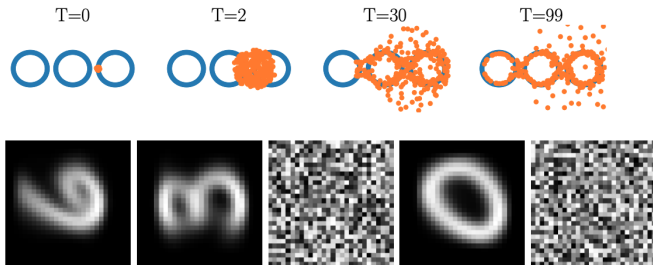


Figure: [BSM25]

- Convergence of MMD flow has been proved under noise injection and stringent conditions on the noise scale [AKSG19, CKK⁺25].
- Disclaimer: There exists many papers where MMD flow **empirically** generate high-quality images with adversarial training of kernels [GBG25], or with non-smooth kernels plus deep network distillation [HWAH24, AHS23].

Convexity and Smoothness

Question: Can we find a new objective $\mathcal{F}$ such that it enjoys **convergence**, similar to $\mathcal{F}_{\text{KL}}$, in the **generative modelling** setting where only samples are available?

Proposition 1 (MMD and χ^2 -divergence)

Suppose μ is absolutely continuous with respect to π , i.e., $\mu \ll \pi$. Then

$$\text{MMD}^2(\mu \parallel \pi) = \left\| T_{\pi}^{\frac{1}{2}} \left(\frac{d\mu}{d\pi} - 1 \right) \right\|_{L^2(\pi)}^2 \quad \text{and} \quad \chi^2(\mu \parallel \pi) = \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)}^2.$$

Here, $T_{\pi} : L^2(\pi) \rightarrow L^2(\pi)$ is the kernel **integral operator** defined as

$$T_{\pi} f(\cdot) = \int k(x, \cdot) f(x) \, d\pi(x).$$

- Similar to $\mathcal{F}_{\text{KL}}$ flow, $\mathcal{F}_{\chi^2}(\cdot) = \chi^2(\cdot \parallel \pi)$ flow enjoys convergence for Gaussian-like π [OT11].
- An **interpolation** between MMD^2 and χ^2 ?

DrMMD: An interpolation of MMD and χ^2 -divergence

Definition (De-regularized Maximum Mean Discrepancy (DrMMD))

Suppose $\mu \ll \pi$ where $\mu, \pi \in \mathcal{P}_2(\mathbb{R}^d)$. Then the (de)-regularized maximum mean discrepancy (DrMMD) between μ, π is defined as, for $\lambda > 0$,

$$\text{DrMMD}(\mu \parallel \pi) = (1 + \lambda) \left\| \left((T_\pi + \lambda \text{Id})^{-1} T_\pi \right)^{\frac{1}{2}} \left(\frac{d\mu}{d\pi} - 1 \right) \right\|_{L^2(\pi)}^2.$$

Proposition 2 (Interpolation of MMD^2 and χ^2)

Suppose k is bounded, continuous, and c_0 -universal.

$$\lim_{\lambda \rightarrow 0} \text{DrMMD}(\mu \parallel \pi) = \chi^2(\mu \parallel \pi), \quad \lim_{\lambda \rightarrow \infty} \text{DrMMD}(\mu \parallel \pi) = \text{MMD}^2(\mu \parallel \pi).$$

- Similar ideas based on spectral regularization have been used for kernel hypothesis testing [MRW⁺99, HBM07, HKSL24].
- Interpolation between MMD and general f-divergences [GAG21, SNRS26].

DrMMD: An interpolation of MMD and χ^2

Question: Does DrMMD inherit the advantages of MMD² and χ^2 ?

- Does $\mathcal{F}_{\text{DrMMD}}$ flow admit **finite sample implementation** of its Wasserstein gradient flow?
- Does $\mathcal{F}_{\text{DrMMD}}$ flow **convergence** when π is Gaussian-like?

Kernel assumptions

Assumption 1 (Growth I)

There exists a constant $M > 0$ such that, for every $x \in \mathbb{R}^d$,

$$\begin{aligned} |k(x, x)| &\leq M (1 + |x|^4), \\ |\partial_{1,i} \partial_{2,i} k(x, x)| &\leq M (1 + |x|^2), \quad i \in [d]. \end{aligned}$$

Assumption 2 (Local Boundedness)

For $R > 0$, let $\mathcal{K} = B(0, R) \subset \mathbb{R}^d$. There exists a constant $B_R > 0$ such that, for every $i, j \in [d]$,

$$\sup_{x \in \mathcal{K}} \|\partial_{1,i} \partial_{1,j} k(x, \cdot)\|_{\mathcal{H}} \leq B_R.$$

- Gaussian, Matérn, and inverse multiquadratic kernels.

Proposition 3 (Finite-sample estimate of the Wasserstein gradient of $\mathcal{F}_{\text{DrMMD}}$)

The Wasserstein gradient of $\mathcal{F}_{\text{DrMMD}}(\cdot) = \text{DrMMD}(\cdot \parallel \pi)$ at μ is $(1 + \lambda) \nabla h_{\mu, \pi}(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, where

$$h_{\mu, \pi} = (T_{\pi} + \lambda I)^{-1} T_{\pi} \left(\frac{d\mu}{d\pi} - 1 \right).$$

Given empirical distributions $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\hat{\pi} = \frac{1}{M} \sum_{i=1}^M y_i$. Given the Gram matrices $K_{xx} \in \mathbb{R}^{N \times N}$, $K_{yy} \in \mathbb{R}^{M \times M}$, $K_{xy} \in \mathbb{R}^{N \times M}$.

$$\begin{aligned} h_{\hat{\mu}, \hat{\pi}}(\cdot) = & \frac{1}{N\lambda} k(\cdot, x_{1:N}) \mathbb{1}_N - \frac{1}{M\lambda} k(\cdot, y_{1:M}) \mathbb{1}_M - \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yx} \mathbb{1}_N \\ & + \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yy} \mathbb{1}_M. \end{aligned}$$

Definition (Poincaré inequality)

We say that π satisfies a **Poincaré inequality** with constant C_P if for all $f, \nabla f \in L^2(\pi)$,

$$\mathrm{Var}_\pi[f] \leq C_P \mathbb{E}_\pi [\|\nabla f\|^2] .$$

- By “Gaussian-like,” we mean that π satisfies a Poincaré inequality, which in particular implies sub-exponential tails [BGL⁺14].
- For example, if $\pi = \mathcal{N}(0, a \cdot \mathrm{Id})$, then $C_P = a^{-1}$.

Proposition 4 (Convergence of $\mathcal{F}_{\chi^2}$ flow)

Suppose that π satisfies the Poincaré inequality. Let $(\mu_t)_{t \geq 0}$ follow the $\mathcal{F}_{\chi^2}$ flow. Then, for any $T \geq 0$, $\mathrm{KL}(\mu_T \| \pi) \leq \exp(-2C_P^{-1}T) \mathrm{KL}(\mu_0 \| \pi)$.

Convergence of DrMMD gradient flow

Theorem 1 (Convergence of DrMMD gradient flow)

Let $(\mu_t)_{t \geq 0}$ be the Wasserstein gradient flow of $\mathcal{F}_{\text{DrMMD}}$. Suppose the target distribution π satisfies a Poincaré inequality with constant C_P . Suppose that $\frac{d\mu_t}{d\pi} - 1 \in \text{Ran}(T_\pi^r)$ with $r > 0$. Then for any $T \geq 0$,

$$\text{KL}(\mu_T \| \pi) \leq \exp\left(-\frac{2(1+\lambda)}{C_P} T\right) \text{KL}(\mu_0 \| \pi) + \mathcal{O}(\lambda^r C_P).$$

- When $\lambda = 0$, it recovers the exponential convergence of χ^2 flow.

$$\text{KL}(\mu_T \| \pi) \leq \exp\left(-\frac{2}{C_P} T\right) \text{KL}(\mu_0 \| \pi).$$

- Larger r means more regularity of the trajectory and thus smaller bias.
- Smaller Poincaré means faster convergence.
- For continuous time DrMMD flow, we would want $\lambda \rightarrow 0$ for ‘convexity’, however, that is not the case for discrete time flow.

Interpolation space $\text{Ran}(T_\pi^r)$

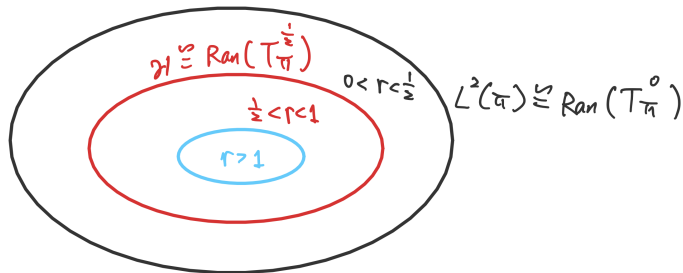


Figure: Visualization of $\text{Ran}(T_\pi^r)$.

- Large r indicates larger smoothness.
- $\text{Ran}(T_\pi^0) \cong L^2(\pi)$ and $\text{Ran}(T_\pi^{\frac{1}{2}}) \cong \mathcal{H}$.

- DrMMD gradient flow (continuity equation)

$$\partial_t \mu_t + \nabla \cdot (\mu_t (1 + \lambda) \nabla h_{\mu_t, \pi}) = 0$$

- DrMMD gradient descent: for step size $\gamma > 0$,

$$\mu_{n+1} = (\text{Id} + \gamma(1 + \lambda) \nabla h_{\mu_n, \pi})_{\#} \mu_n.$$

Convergence of DrMMD gradient descent

Proposition 5 (Descent lemma of DrMMD gradient descent)

Let $(\mu_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose $\pi \propto \exp(-V)$ with $\mathbf{H}V \preceq \beta \mathbf{I}$. Suppose all assumptions in the proposition of the PL condition on $\mathcal{F}_{\text{DrMMD}}$ hold. Suppose the step size γ is small enough.

$$\text{KL}(\mu_{n+1} \parallel \pi) - \text{KL}(\mu_n \parallel \pi) \leq -\frac{2}{C_P} \chi^2(\mu_n \parallel \pi) \gamma + \underbrace{\gamma \lambda^r}_{\text{Approximation error}} + \underbrace{\gamma^2 \beta \chi^2(\mu_n \parallel \pi) \frac{1}{\lambda}}_{\text{Discretization error}}$$

- A **trade-off** between the approximation error and the time-discretization error.
- Optimal choice of **adaptive** λ_n at each iterate n :

$$\lambda_n \propto \chi^2(\mu_n \parallel \pi)^{\frac{1}{r+1}}$$

- At the start, we want a **larger** λ to have more smoothness; when closer to the convergence, we want a **smaller** λ to operate in the χ^2 regime to better catch the difference.

Convergence of DrMMD gradient descent

Theorem 2 (Convergence of DrMMD gradient descent)

Let $(\mu_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose all conditions from the descent lemma hold. Then, for any $n_{\max} \in \mathbb{N}$,

$$\text{KL}(\mu_{n_{\max}} \parallel \pi) \leq \exp\left(-\frac{2n_{\max}\gamma}{C_P}\right) \text{KL}(\mu_0 \parallel \pi) + \mathcal{O}(\gamma^{\frac{r}{r+1}} C_P \beta^{\frac{r}{r+1}}).$$

- To reach error $\text{KL}(\mu_{n_{\max}} \parallel \pi) \leq \delta$, it takes $\mathcal{O}((\frac{1}{\delta})^{\frac{r+1}{r}} \log \frac{1}{\delta})$ iterations.
- By comparison, for Langevin Monte Carlo, it takes $\mathcal{O}(\frac{1}{\delta} \log \frac{1}{\delta})$ [CEL⁺24].
- DrMMD gradient descent takes more iterations due to the additional approximation error $\mathcal{O}(\lambda^r)$, but it **operates without the knowledge of potential V** and only requires $\Sigma_\pi = \int k(x, \cdot) \otimes k(x, \cdot) d\pi$ and the embedding $\int k(x, \cdot) d\pi$.

Conclusions

- We propose DrMMD gradient flow as an interpolation of MMD gradient flow and χ^2 gradient flow.
- DrMMD gradient flow / descent has **global convergence** results, compared to MMD flow, under an adaptive regularization parameter λ .
- This justifies the application of **adaptive kernels** in recent MMD flow (MMD GAN) papers that achieve SOTA empirical performances.
- More in the paper <https://arxiv.org/pdf/2409.14980>.
 - Empirical results on synthetic datasets.
 - An example of DrMMD flow with a Gaussian target distribution π which satisfies all conditions in the theorems.
 - Finite-particle convergence results with propagation of chaos bound.

About Me



- Zonghao Chen
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